Ecological Processes

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A network meta-analysis on responses of forest soil carbon concentration to interventions

Kaihui Shen[1], Lin Li[1*], Shiguang Wei[2], Jiarun Liu[1] and Yi Zhao[1]

Abstract

Background Forests play a crucial role in absorbing CO_2 from the atmosphere. 55% of the carbon in terrestrial ecosystems is stored in forests, with the majority of forest carbon stored in soil. To better understand soil organic carbon (SOC) of forests and to access interventions that affect their SOC concentration, we conducted a comparative analysis between natural and planted forests. Forest interventions refer to the actions taken by humans to manage, protect, or transform forests, and can be divided into two main categories: environmental intervention and anthropogenic intervention. This study focused on the effects of different interventions on SOC in natural and planted forests by reviewing a total of 75 randomized controlled trials in the global literature and extracting a total of 15 different interventions.

Results Through network meta-analysis, we found that natural forests have 22.3% higher SOC than planted forests, indicating their stronger carbon storage function. In natural forests, environmental interventions have a stronger impact. SOC is significantly influenced by forest age, fertilization, and elevation. In planted forests, however, anthropogenic interventions have a stronger impact. Pruning branches and fertilization are effective interventions for planted forests. Furthermore, forest degradation has a significantly negative impact on SOC in planted forests.

Conclusion Overall, interventions to enhance soil carbon storage function differ between natural and planted forests. To address global climate change, protect biodiversity, and achieve sustainable development, it is imperative to globally protect forests and employ scientifically sound forest management practices. Regarding natural forests, the emphasis should be on comprehending the effects of environmental interventions on SOC. Conversely, concerning planted forests, the emphasis should be on comprehending the effects of anthropogenic interventions.

Keywords Natural forest, Planted forest, Soil organic carbon (SOC), Network meta-analysis, Intervention

Background

Restoring forest carbon storage is a goal of "the UN Decade on Ecosystem Restoration" (Hua et al. 2022). Forest carbon storage is an important climate change mitigating measure (Rossi et al. 2009). Enhancing the function of forest carbon storage is necessary to sooner achieve "net-zero" CO_2 emissions. Forests can absorb 31% of anthropogenic CO_2 emissions (Piao et al. 2022). Forest ecosystems are the core of terrestrial ecosystems (Suhaili et al. 2021), with forest soil storing approximately 1500–2000 Pg carbon, which is 1.4–3.8 times more than plants and 8.0–22.8 times more than forest

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litter (Rossi et al. 2009). Globally, the forest soil pool accounts for about 40% of the total soil carbon pool (Rossi et al. 2009). Therefore, the organic carbon of forest soils plays a critical role in ecosystem carbon balance and sensitively reflects changes in forest carbon storage (Nave et al. 2013).

Soil organic carbon (SOC) reflects soil fertility and quality, playing a key role in the sustainable development of forest ecosystems. Forests include natural forests (restored and preexisting native forests) and planted forests (formed through human tree planting, cultivation, and management) (Fonseca et al. 2022). However, with increasing climate change and natural disturbances, carbon storage potential of forests has declined. In response to this change, scholars are attempting to address it by adapting management practices in forests that are currently below their natural equilibrium biomass level (Erb et al. 2018) and expanding forest areas through afforestation and reforestation programs (Roebroek et al. 2023). Forest interventions refer to the actions taken by humans to manage, protect, or transform forests, and can be divided into two main categories: environmental intervention and anthropogenic intervention. Environmental intervention involves adjusting the natural environment to influence the evolution and development of the forest ecosystem, typically achieved through natural processes. Anthropogenic intervention, on the other hand, consists of measures directly taken by humans to achieve specific forest management goals, such as logging, pruning, fertilization, and other practices. Studying the effects of various interventions on SOC in natural and planted forests can not only assess their carbon storage potential but also help to adopt more effective measures to enhance soil productivity and promote soil carbon storage function.

Various interventions also have different effects on SOC in the natural and planted forests. In natural forests, the following interventions affect SOC: selective logging, wildfires, overhunting (Berenguer et al. 2014), burning (Fonseca et al. 2022), forest thinning (Suhaili et al. 2021), and forest logging residues (Souza et al. 2020). In planted forests, many interventions can affect SOC, such as fertilization (Barros et al. 2021), thinning (Gross et al. 2018), fire management (Lucas-Borja et al. 2021), and so on. The aboveground and underground parts of the forest ecosystem are interconnected. Plants provide litter and root exudates as organic matter to soil organisms, while decomposers in the soil provide mineralized nutrients to plants (Fan et al. 2015). These interventions can directly or indirectly influence carbon storage capacity of soils and contribute to climate change mitigation. However, our understanding of the relative importance of these different factors in influencing soil carbon storage is poor due to a lack of comprehensive quantitative research data on a large scale.

Meta-analysis is commonly used to consolidate scientific understanding and identify sources of variation in independent research findings. Some scholars have conducted comprehensive meta-analyses to extract and analyze big data to determine the main factors and their effects on forest carbon sequestration on a large scale (Guo and Gifford 2002; Chen et al. 2015; James and Harrison 2016; James et al. 2021). Currently, factors that have been identified as influencing SOC include climate, land use type, forest age, forest type, management practices, and nitrogen deposition (Barcena et al. 2014). Existing meta-analyses have examined the effects of various interventions on SOC in natural or planted forests in specific regions. These studies have found that SOC increases with elevation (Tashi et al. 2016), selective logging and wildfires have minimal effects on SOC in tropical rainforests (Berenguer et al. 2014), the original land use type is a major factor affecting SOC changes after afforestation in Northern Europe (Barcena et al. 2014), and plant root exudates rapidly affect the decomposition of SOC (Yan et al. 2023). While these studies significantly enhance our understanding of the effects of various interventions on forest organic carbon or carbon storage, they lack comprehensive assessments of broad-scale interventions on natural and planted forest sinks.

In this study, we conducted a network meta-analysis (NMA) to comprehensively analyze a large amount of available literature on forest SOC. We extracted and analyzed 75 of the most relevant published studies, which provided 242 observed results. The aim was to assess and rank the direct and indirect effects of different interventions on forest SOC. Specifically, our study aimed to (1) compare the differences in soil carbon storage function between natural forests and planted forests based on existing global research; (2) elucidate the current impact of various interventions on SOC in natural forests to identify the most effective interventions for promoting SOC; (3) understand the current impact of various interventions on SOC in planted forests to identify the most effective interventions for promoting SOC.

Materials and methods

Data compilation

In this study, we conducted a global systematic review and comparison of SOC concentration in natural forests (referring to forests that are essentially maintained in their natural state with minimal human interference) and planted forests (referring to forests formed through human tree planting), cultivation, and management activities, where the vegetation consists mainly of tree

species introduced through anthropogenic intervention (Veen et al. 2010; Chazdon et al. 2016). We searched the Web of Science (http://apps.webofknowledge.com/) and the China National Knowledge Infrastructure (https://www.cnki.net/) electronic database from the date of their inception to April 8, 2023, with no language restrictions. We used the search terms 'forest soil organic carbon' or 'forest soil carbon' and 'controlled trial' or 'randomized controlled trial (RCT)'. Two researchers independently screened the articles based on the inclusion criteria, with consensus reached through discussion. In cases of disagreement, an independent expert was consulted for his opinion. The specific literature screening process and inclusion criteria are outlined in Fig. 1.

Following these criteria, our database comprises two datasets, namely, natural forests and planted forests. To ensure the validity of the study results, we selected data extraction from randomized controlled trials (RCT) of high research quality as reported in the literature. The names of the extracted interventions were based on the literature, and the usage level of interventions was not differentiated throughout the entire text. These datasets include SOC concentration (including sample size, mean, standard deviation, and standard error) from forest plots that were subjected to different interventions and left untreated as controls. These datasets are utilized for analysis and evaluation to investigate the impact of different interventions on SOC. When data in the publications were tabulated, we directly extracted it; however, when presented graphically, we employed the Get Data software for extraction.

In the end, our database for this study was compiled from 75 studies in a network meta-analysis. Specifically, there were 35 studies on natural forests and 40 studies on plantations. The natural forest studies included 118 observations from 29 sites, while the plantation studies included 124 observations from 36 sites. The distribution of sampling sites can be seen in Fig. 2. Field experiments for data extraction were conducted across five continents and involved 19 countries. We included a total of 15 interventions, among which environmental interventions consist of acid rain (AR), elevation (ASL), CO_2 concentration (CC), forest age (FA), forest degradation (FD), flooding (FL), and temperature (TEMP); and anthropogenic interventions consist of nitrogen addition (AN), sulfur addition (AS), biological disturbance (BD), burning litter (BL), fertilization (F), fire severity (FS), forest thinning (FT), and trimming branches (TB).

Data calculation

Although most reports of organic carbon concentration are in units of $g \cdot kg^{-1}$, for some data presented as soil organic carbon content, we may convert it using soil bulk density or soil moisture content data. We utilized a conversion factor of 0.58 to convert as soil organic matter (SOM) to SOC concentration (Bao 2010). Due to variations in sampling depth among different experiments, which can affect the qualitative assessment of SOC concentration, we compiled sample data from different layers of soil profiles into one dataset. The average SOC concentration was then used for the entire soil profile (Liao et al. 2012). Because variations in sampling depths between experiments would not significantly affect the qualitative assessment of SOC concentration. Therefore, the average SOC concentration along the entire soil profile was used throughout (Liao et al. 2012).

To characterize the ability of forest soil to sequester carbon, we chose the primary outcome was SOC concentration, measured by comparing the SOC concentration before any intervention and after each specific

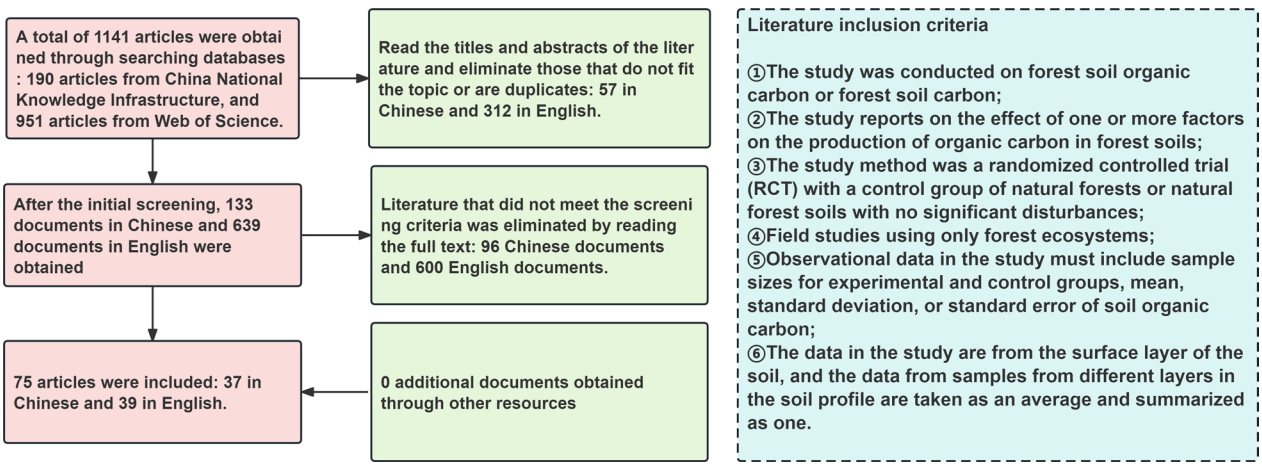


Fig. 1 Literature screening process diagram

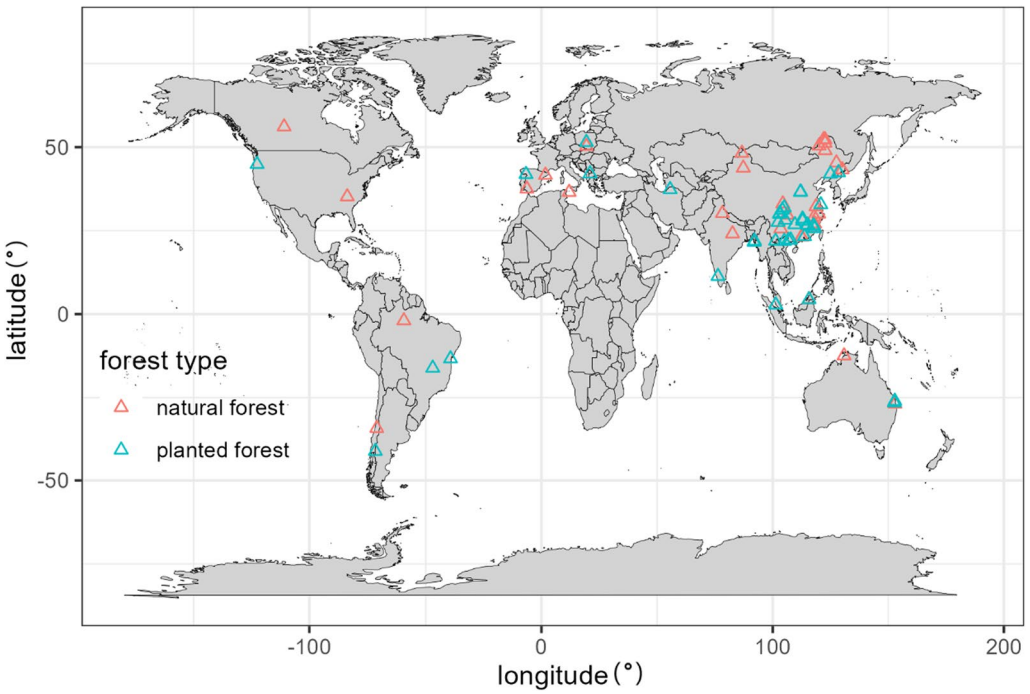


Fig. 2 Distribution of forest soil sample sites extracted in this study

intervention. This comparison allows us to assess the extent to which each intervention influences the soil organic carbon concentration. We estimated standardized mean differences (SMD, Cohen's *d*) for continuous outcomes using pairwise and network meta-analysis. In addition, in the network meta-analysis, we assumed that the amount of heterogeneity was the same for all treatment comparisons. To assess the amount of heterogeneity, we compared the posterior distribution of the estimated heterogeneity variance with its predictive distribution. To rank the treatments for each outcome, we used the surface under the cumulative ranking curve (SUCRA) and the mean ranks. We performed a statistical assessment of consistency (i.e., the agreement between direct and indirect evidence) using the design-by-treatment test and by separating direct and indirect evidence (Dias et al. 2010; Higgins et al. 2012).

Data analysis

We conducted a network meta-analysis, which is a systematic review method that forms a network structure by directly and indirectly comparing the effects among multiple different interventions, providing a more comprehensive assessment of intervention effects.

We conducted all statistical analyses utilizing Excel, R (version 4.1.3) and STATA16. R (version 4.1.3) was used for various tasks such as one-way analysis of variance (ANOVA) followed by the least significant difference (LSD), effect size calculation, setting network attributes, creating network plots, setting up and running Bayesian models, generating global site maps, and creating various forest plots. The software packages utilized include the 'multinma' package, 'rstan' package, and 'ggplot2' package. In addition, STATA16 software with the 'Network' package is employed specifically for calculating and ranking the SUCRA values of different interventions. The closer the SUCRA value is to 1, the higher the intervention ranks, indicating its relatively better effectiveness. Conversely, a SUCRA value closer to 0 suggests that the intervention ranks relatively lower. Visualization of the data was achieved using R (version 4.1.3), STATA16, and Origin 2021.

Results

The soil carbon sink functions of natural and planted forests

Both natural forests and planted forests were studied separately to determine their SOC concentration. The results showed a significant difference ($P < 0.05$) in SOC concentration between natural forests and planted forests (Fig. 3). SOC concentration in natural forests ranges from 6.8 to 140.3 g·kg^{-1}, with an average of 36.2 g·kg^{-1}; whereas in planted forests, it ranges from 2.74 to 148.3 g·kg^{-1}, with an average of 29.6 g·kg^{-1}. Specifically, SOC concentration in natural forests was found to be approximately 22.3% higher than that in planted forests.

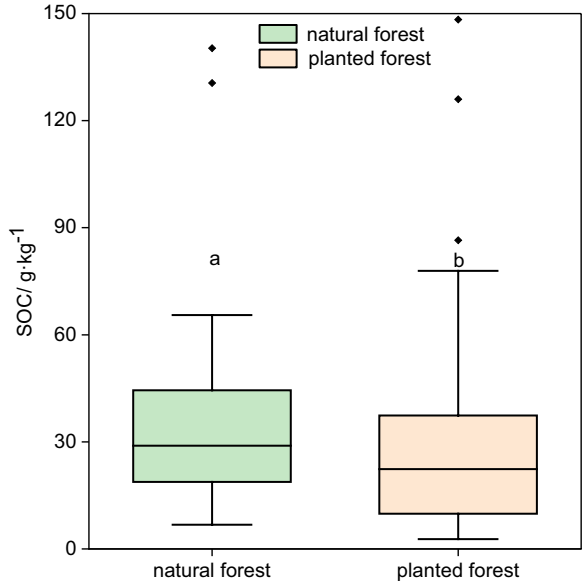


Fig. 3 Comparison of SOC between natural and planted forests. The "a" and "b" in the figure indicate significant differences in SOC between natural forests and planted forests at the 0.05 level

The SOC concentrations of both natural and planted forests exhibit an exponential distribution (Fig. 4). Compared to natural forests, SOC concentration decreases more rapidly in planted forests, but with a greater vertical offset. The highest frequency of SOC concentration in natural forests occurs between 20 and 40 g·kg^{-1}, with no distribution observed between 80 and 120 g·kg^{-1}. In contrast, the highest frequency of SOC concentration in planted forests occurs between 0 and 20 g·kg^{-1}, with no distribution observed between 100 and 120 g·kg^{-1}.

The effects of different interventions on soil carbon storage in natural forests and planted forests, and their importance ranking

This study conducted separate statistical analyses using undisturbed natural and planted forests as control groups, and the evidence from the network meta-analysis is presented in Fig. 5. Among the studies that included 13 different interventions with undisturbed natural forest soil as the control group, a consistency model analysis ($P=0.161>0.05$) showed that only AN and AS formed a closed-loop consistency with the natural forest soil control group. Notably, AR, FS, and AN had the higher number of studies. An evidence network was also constructed for the effects of 14 different interventions on SOC in planted forests. Using a consistency model analysis ($P=0.065>0.05$), it was found that TEMP, AN, BL, BD, and F were found to form a closed loop with planted forests (PF). It is noteworthy that among all the studies, interventions with the higher number of studies included AN, FT, BL, BD, F, FS, and TEMP.

Various interventions have different effects on SOC in natural forests and planted forests, as depicted in Fig. 6. Specifically, in natural forests, FA, F, and ASL significantly increase SOC with 95% confidence intervals (CI) values ranging from 4.22 to 9.62, 4.53 to 8.44, and 0.22 to 1.31, respectively. In addition, AN, AR, AS, FT, and TB also have a positive effect. On the other hand, BL, FD, FL, FS, and TEMP have a negative effect on SOC in natural forests. The effects of different interventions on SOC in planted forests are as follows: TB and F had significant positive effects on SOC with 95% CI intervals of 0.03–3.13 and 0.35–1.20, respectively. In addition, AR, BD, BL, CC, FT, and FA also had a positive effect on SOC in planted forests. Whereas, FD had a significant

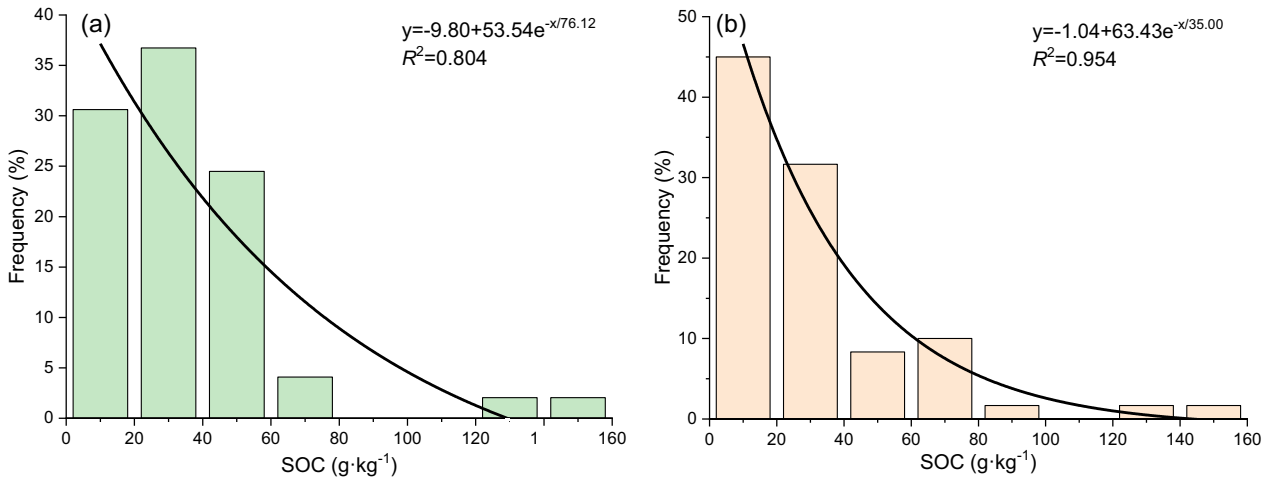


Fig. 4 Frequency distribution plot of SOC concentration in natural (**a**) and planted (**b**) forests

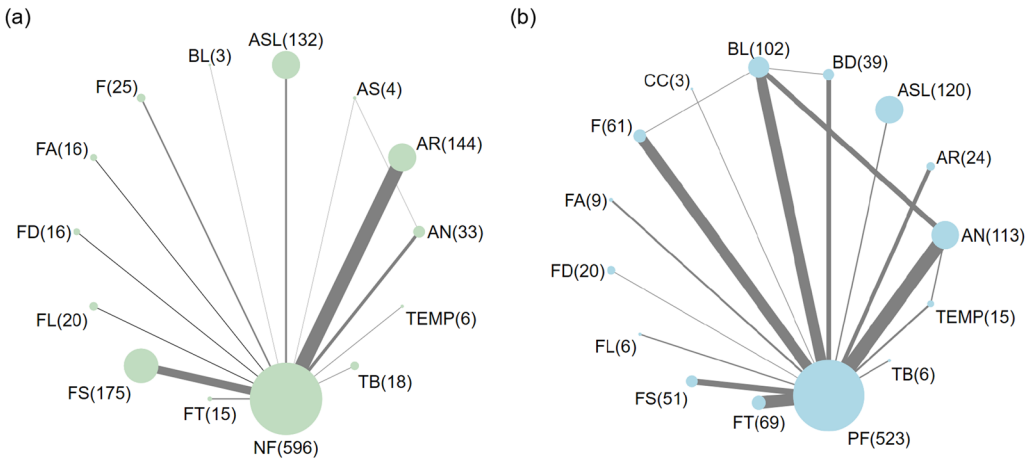


Fig. 5 The network meta-analysis evidence diagram of interventions included in the study in the (**a**) natural forest and (**b**) planted forest. AN: nitrogen addition; AR: acid rain; AS: sulfur addition; ASL: elevation; BD: biological disturbance; BL: burning litter; CC: CO_2 concentration control; F: fertilization; FA: forest age; FD: forest degradation; FL: flooding; FS: fire severity; FT: forest thinning; NF: natural forest; PF: planted forest; TB: trimming branches; TEMP: temperature. Width of the lines is proportional to the number of trials comparing every pair of treatments. The size of every circle is proportional to the number of randomly assigned sample sizes

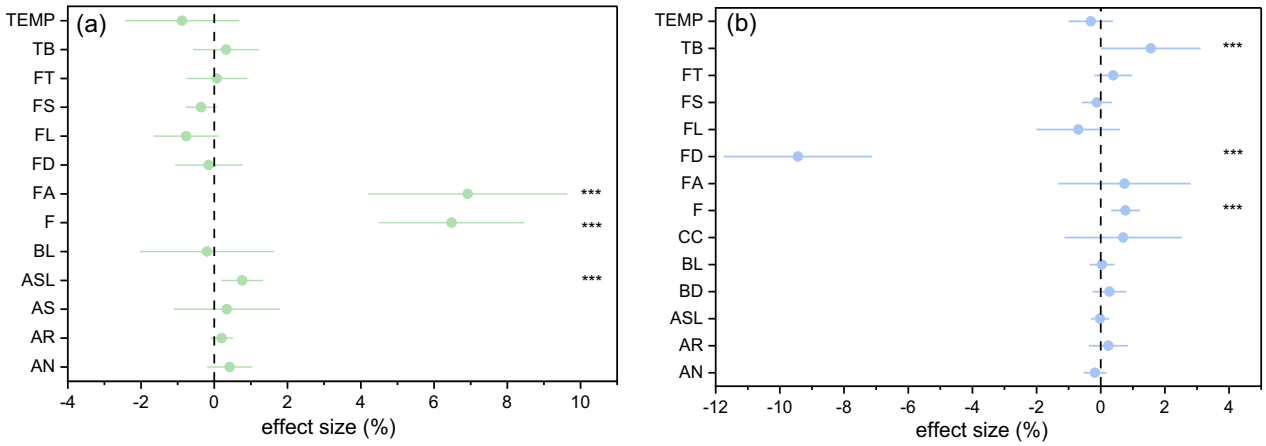


Fig. 6 The impact values of different interventions on SOC in (**a**) natural and (**b**) planted forests. The average effect size of SOC in natural forests, under various interventions, is represented with error bars indicating the 95% confidence intervals for the weighted percentage change. The asterisks ***indicate that the interventions have a statistically significant impact on SOC ($P < 0.05$). The abbreviations for each intervention are the same as shown in Fig. 5

negative effect on SOC with a 95% CI interval from − 11. 73 to − 7.15. AN, ASL, FL, FS, and TEMP also had a negative effect on SOC in planted forests.

The overall efficiency SUCRA probability ranking of the effect of interventions on SOC in natural and planted forests is shown in Fig. 7. In natural forests, the 13 interventions are ranked in descending order as follows: FA, F, ASL, AN, TB, AS, AR, FT, BL, FD, FS, TEMP, and FL. Among them, there are 8 interventions with SUCRA values greater than those of control group and 5 interventions with values lower than those of control group. In planted forests, the 14 interventions are ranked in descending order as follows: TB, F, CC, FA, FT, BD, AR, BL, ASL, FS, AN, TEMP, FL, and FD. Among these, there are 8 interventions with SUCRA values greater than those of control group and 6 interventions with values lower than those of control group. A SUCRA value greater than that of the control group indicates that the corresponding intervention is relatively better than no intervention. Specifically, a higher SUCRA indicates that the intervention is relatively more effective in increasing SOC.

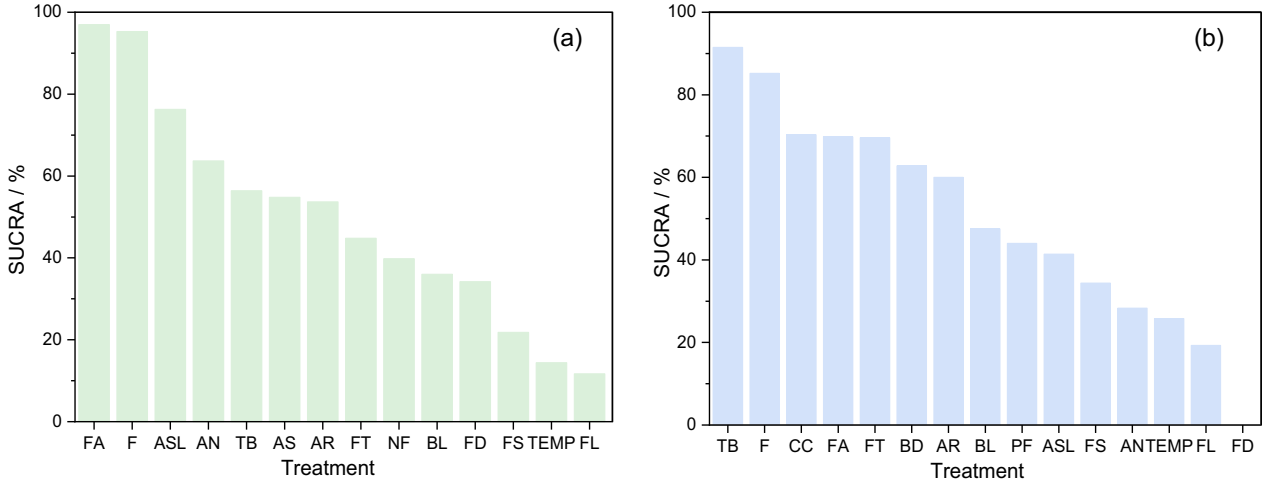


Fig. 7 The ranking of the effects of various interventions on SOC of (**a**) natural and (**b**) planted forests. The bar represents the SUCRA values of each intervention, and the SUCRA values of each intervention decrease in order from left to right. The abbreviations for each intervention are the same as shown in Fig. 5

Discussion

Differences in soil carbon storage functions between natural and planted forests

Planted and natural forests have significant differences in soil carbon storage (Freier et al. 2010). A quantitative review found that planted forests have experienced an 18.7% reduction in soil carbon pool compared to natural forests on terrestrial land, excluding Antarctica (Liao et al. 2012). An earlier study also found a 13% reduction in the soil carbon pool when natural forests were converted to planted forests. Converting natural tropical forests to rubber plantations results in a loss of 10 mg of carbon per hectare of land (Guillaume et al. 2015). Our comprehensive comparison revealed that planted forests have a 22.3% lower SOC concentration than natural forests (Fig. 3). As planted forests have higher net primary productivity, aboveground biomass, litter, and fine root biomass than natural forests, this consequently leads to a lower soil carbon input in planted forests than in natural forests. The structure of carbon storage varies with geographical location and forest type; tropical forests store 56% of carbon in biomass and 32% in soil, while northern forests store only 20% in biomass but up to 60% in soil (Pan et al. 2011). Anthropogenic disturbance can disrupt the original soil structure, alter soil microbial communities, and weaken the role of microbes in carbon storage (McLeod et al. 2011). In general, forests with low levels of disturbance have higher SOC concentrations. Naturally forested areas with minimal disturbance tend to have higher SOC concentrations than highly disturbed planted forests.

Key interventions to promote soil carbon storage functions in natural forests

The results of the network meta-analysis indicate that in natural forests, the effectiveness of interventions on SOC are ranked from highest to lowest as follows: FA, F, and ASL. The effect of environmental interventions is stronger than that of anthropogenic interventions (Figs. 6a and 7a).

Forest age has the greatest impact on SOC in natural forests. Some studies have indicated a pattern of increasing SOC with the age of natural forests on a global scale. SOC of *Quercus mongolica* forests in northeast China increases with forest age. The development of the forest leads to a greater input of organic matter into the soil, and the increased activity of soil microorganisms also contributes to the accumulation of SOC (Xu et al. 2019). Research on secondary forests in central Panama has also revealed that secondary forests older than 40 years have higher SOC concentration than surrounding younger secondary forests (Neumann-Cosel et al. 2011). Moreover, the initial 10–30 years of forest succession have the greatest impact on the recovery of SOC (Batterman et al. 2013; Berenguer et al. 2014). However, there is also research indicating that the SOC of older forests declines, maybe because with increasing forest age, SOC is lost in the form of carbon dioxide, while nitrogen and phosphorus are returned to the soil in a stable form through nutrient cycling, leading to changes in soil stoichiometry (Fan et al. 2015; Zhang et al. 2019).

Fertilization is the second major intervention affecting SOC in natural forests (Fig. 7a). However, the type of fertilizer, the method of application, and the timing

of fertilization can have different effects on SOC. The application of nitrogen and phosphorus fertilizer can influence soil stoichiometry, resulting in changes in SOC (Fan et al. 2015; Zhang et al. 2019). Research has indicated that the application of organic fertilizers can enhance soil microbial activity and microbial biomass (Chu et al. 2007). Adding organic fertilizers, such as glucose, cellulose, simple organic compounds, forest litter, straw, and biochar to the soil, has various effects on soil microbes, enzymes, and the active carbon pool. In the short term, these additions can change the structure of microbial communities, soil enzyme activity, and the amount of active organic carbon (Liu et al. 2022). Over time, some of this added carbon will be decomposed by soil organisms into simple inorganic compounds such as CO_2 and H_2O. Another portion will undergo carbon chain breakdown and recombination under microbial action to form more complex and recalcitrant compounds (Lu et al. 2014).

Elevation is another important factor influencing SOC in natural forests (Fig. 7a). Research conducted in the eastern Himalayas revealed that there is a notable increase in total soil carbon with increasing elevation (Tashi et al. 2016). This can be attributed to changes in abiotic factors such as temperature, humidity, and solar radiation that affect forest composition and the accumulation of SOC (Jobbagy and Jackson 2000; Laughlin and Abella 2007; Singh et al. 2011; Simon et al. 2018). Higher elevations are associated with lower temperatures, which affect both the input and decomposition of SOC. While production and decomposition rates decrease, the proportional decrease in decomposition rate is relatively small, resulting in higher SOC at higher elevations (Tashi et al. 2016). Temperature also influences metabolic processes associated with carbon acquisition by plants and microorganisms through photosynthesis and respiration. Consequently, this affects metabolism, production rates, distribution patterns, carbon balance, and mortality rates (McLeod et al. 2011). In addition, studies also suggest that biological interactions become less intense with increasing elevation (Zvereva and Kozlov 2022), leading to reduced soil disturbance and increased SOC. Therefore, protecting natural forests from human-induced disturbance is an effective way to promote forest carbon sinks. Natural forests at higher elevations have a stronger carbon storage function than those at lower elevations. In addition, applying appropriate fertilization can increase the concentration of active organic carbon in the soil, thereby enhancing carbon storage function of forest soils.

Key interventions to promote soil carbon storage function in planted forests

SOC in planted forests is primarily influenced by anthropogenic interventions such as pruning and fertilization (Figs. 6b and 7b). Human disturbances strongly impact the amount of aboveground litter that enters the soil (Miao et al. 2019). Improper land management can lead to soil carbon depletion and reduced carbon storage capacity. SOC loss is a major contributor to soil degradation (Fonseca et al. 2022). Tree pruning has dual effects on forest ecosystems. On one hand, it stimulates tree regeneration and promotes the growth of new branches and trees, which absorb CO_2 and contribute to an increase in SOC storage in the forest (Chambers et al. 2013). On the other hand, branch pruning can lead to the accumulation of plant residues such as branches, leaves, and bark on the forest floor. These residues contribute to the SOC of the area (Houghton 2005).

Fertilization is an important factor influencing soil fertility, particularly in planted forests. It can impact key indicators of soil fertility such as pH, carbon, and nitrogen. Even minor changes in these factors can affect forest carbon cycling (Chen et al. 2015). Nitrogen fertilizer can effectively stimulate the microbial decomposition of recalcitrant organic matter such as humus or lignin-containing organic matter. As a result, more carbon released into the environment (McLeod et al. 2011). In tropical secondary forests, there is a strong correlation between soil nitrogen concentration and soil carbon storage over time (Jones et al. 2019). While some studies indicate that adding nitrogen does not affect overall soil carbon storage, it does increase the concentration of soluble organic carbon within the soil (Chen et al. 2015). Under low nitrogen conditions, plant root exudates stimulate microbial activity and enzyme production, which promotes the decomposition of SOC (Ma et al. 2023). Long-term increases in SOC following afforestation indicate that forest restoration can significantly contribute to global carbon sequestration (Rossi et al. 2009). Therefore, implementing effective management practices, such as pruning and fertilization, can further increase the capacity of planted forests to sequester soil carbon.

Conclusion

This article compared SOC concentration between natural forests and planted forests and conducted a systematic review through network meta-analysis to examine the effects of various interventions on SOC in both natural and planted forests. The findings are as follows: (1) Natural forests have higher SOC concentration than planted forests. (2) SOC in natural forests is significantly influenced by environmental interventions, with factors

such as forest age and elevation playing an important role. Fertilization is also an effective intervention for SOC in natural forests. (3) SOC in planted forests is significantly influenced by anthropogenic interventions, with practices such as pruning and fertilization having a significant impact on SOC concentration. However, forest degradation can have a significant negative effect on SOC in planted forests.

Global efforts are needed to increase carbon storage capacity of forest soils. Protecting natural forests from disturbance and increasing stand age can improve soil carbon storage capacity. Forests at higher elevations tend to exhibit stronger carbon storage capabilities. In addition, appropriate fertilization can also enhance this function. Appropriate human management (pruning and fertilization) of planted forests can enhance their soil carbon storage capacity and promote carbon accumulation in planted forest soils.

Acknowledgements
Not applicable.

Author contributions
Kaihui Shen, Lin Li, Shiguang Wei contributed to the conception of the study; Kaihui Shen, Lin Li, Shiguang Wei, Jiarun Liu contributed significantly to literature retrieval, literature screening and data extraction. Kaihui Shen, Lin Li, Shiguang Wei, Jiarun Liu, Yi Zhao performed the data analyses and wrote the manuscript; Kaihui Shen, Lin Li, Shiguang Wei helped perform the analysis with constructive discussions.

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Availability of data and materials
The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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